

3D MODEL · SPEC CONVERSION

SP3D Piping Spec Auto Generator from PMC

Analyzes a piping-spec PDF and auto-generates a SmartPlant 3D reference-data spec workbook (HanaMaster catalogue based).

Key Purpose

Why this program

1. Auto-build the SP3D Reference-Data spec workbook

– Parses the PMC piping-material classes straight out of the MPS PDF into an SP3D spec .xlsx

→ **Effect:** The reference-data workbook is generated in minutes instead of days of manual data entry, with no transcription error

2. Validate every commodity code against the catalogue

– Each code from the MPS is checked against the HanaMaster catalogue folder before it is written

→ **Effect:** Invalid or missing parts are caught up front, so the spec loads cleanly into SP3D instead of failing on import

3. Regression-check against the r0 baseline

– Validate vs r0 re-runs the class data and reports differences from the approved revision

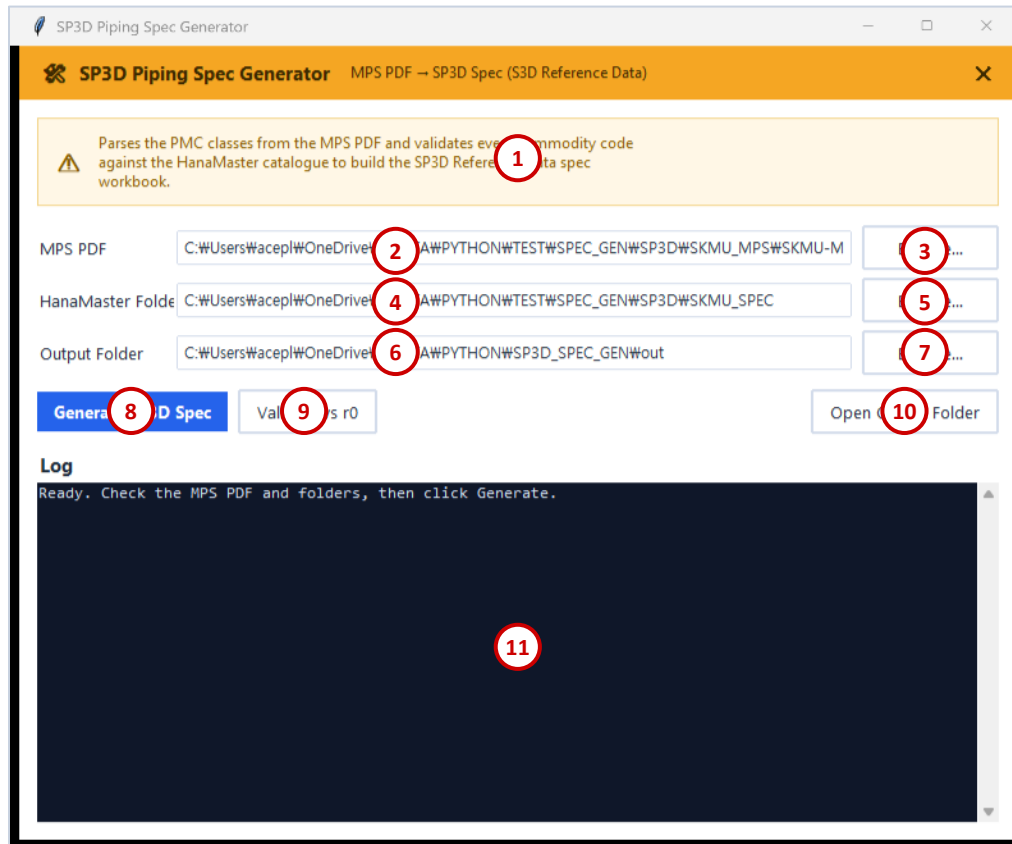
→ **Effect:** Changes between revisions are surfaced automatically, giving confidence the workbook stays consistent

4. Fully offline and repeatable

– Pick PDF + catalogue + output folder, click once; the whole build runs locally with a live log

Required inputs : MPS PDF + HanaMaster catalogue folder + Output folder

Main Window – Item Guide



1. Purpose banner — parse PMC classes & validate codes
2. MPS PDF path (source spec PDF)
3. Browse... for MPS PDF
4. HanaMaster catalogue folder path
5. Browse... for HanaMaster folder
6. Output folder path
7. Browse... for output folder
8. Generate SP3D Spec (build workbook)
9. Validate vs r0 (regression check)
10. Open Output Folder
11. Log — progress & results

Overview

What it does

1. Analyzes a piping-spec PDF and auto-generates a SmartPlant 3D reference-data spec workbook (HanaMaster catalogue based)

Category: 3D Model · Spec Conversion

Folder: SP3D_SPEC_GEN

Main scripts: src/app.py

Scripts / Status: 13 scripts · Operational

Tags: SP3D · SPEC · Excel · PDF · catalog

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: 3D Model · Spec Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: src/app.py.
- Output samples and the program window above reflect the current build.

Thank You

SP3D Piping Spec Auto Generator from PMC · Plant Engineering Total Service System